Feasibility First: Auditing Reference Dynamics Before Adaptive Humanoid Motion Tracking

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Abstract— **Failure-adaptive sampling is intended to concentrate humanoid tracking updates on motions the policy has not mastered. Policy error, however, can conflate controllable difficulty with a reference that demands unsupported or over-limit robot dynamics. In a three-seed Unitree G1 campaign, peak top-1 allocation reached 87–89%, and the same kneel-and-crawl reference became a dominant attractor in every seed. During that reference's descent, 86% of frames have no collision geometry within 6 cm of the floor, and median unsupported force is approximately one body weight. We introduce a policy-independent screen that applies contact-free inverse dynamics and a torque-limited contact program to the final robot-space trajectory before training. On one AMASS-to-G1 pipeline, the screen flags 2,442 of 10,705 clips above a fixed 10%-of-frames threshold; on a separate filtered 4,950-clip production pairing, an independent implementation flags 7, and the implementations agree on 39 of 40 strict decisions in a stratified same-clip panel. We convert feasible intervals into 1,184 hash-bound units with 368,951 legal fixed-horizon starts, enabling a controlled comparison of learning-progress and deployment-uniform allocation on identical support. [Phase G pending: endpoint access remains locked.] The present evidence supports an embodiment- and pipeline-specific release audit, not a generic prevalence rate, a hardware-safety certificate, or a claim that filtering improves policy performance.**

I. Introduction

Large retargeted motion banks have made generalist humanoid tracking a practical training recipe: map human motion to a robot, train a policy over the resulting references in massively parallel simulation, and spend additional updates on motions the policy currently fails. Systems such as PHC, MaskedMimic, H2O, ExBody, BeyondMimic, and SONIC demonstrate the scale and capability of this recipe [1], [2], [3], [4], [5], [6]. Failure, completion, or recent learning progress appears to tell the trainer where another update is most useful [5], [7], [8].

That interpretation assumes that policy error measures a solvable control problem. A retargeted trajectory may instead have no admissible contact capable of supplying its demanded base wrench, or require joint effort beyond the modeled actuator range. The same policy failure then admits two incompatible readings: the controller has not learned a feasible skill, or the reference does not define a feasible skill for this embodiment and scene.

We observed this ambiguity in a 100-motion Unitree G1 campaign. A BeyondMimic-style sampler concentrated on the same kneel-and-crawl clip in all three seeds. Peak top-1 mass was 0.884, 0.870, and 0.893; those campaign maxima belonged to another clip in two seeds. Clip-uniform sampling nevertheless achieved higher held-out survival in all three paired seeds. The attractor passes ordinary kinematic checks. Yet from 0.75 to 1.75 s its pelvis descends from 0.79 to 0.40 m while the nearest collision geometry remains 7–10 cm above the floor. A median 329 N of support demand, approximately the 327 N model weight, has no modeled source.

Three ambiguities must be removed before testing adaptive allocation. *Error ambiguity* mixes policy competence with reference feasibility. *Support ambiguity* arises when clip-level sampling changes the duration prior or admits fixed-horizon starts that cross invalid frames. *Evaluation ambiguity* arises when wraparound, implicit teleports, reset-state terminal reads, or unstable attribution make unlike trials appear paired.

Our interface has two parts. A robot-space dynamic-feasibility screen analyzes the final retargeted trajectory using the target MuJoCo model, modeled contacts, friction, and actuator ranges. An exact-support contract turns feasible intervals into non-wrapping fixed-horizon starts, stable attribution units, and paired evaluation conditions bound by content hashes. The allocation rule may then change while embodiment, reward, PPO, support, legal-start prior, caps, compute, seeds, and evaluator remain fixed.

The evidence has an important boundary. The screen flags 2,442/10,705 clips in one AMASS-to-`whole_body_tracking`-to-G1 pipeline, but only 7/4,950 in a separately filtered BONES-SEED-to-G1 production pairing used by SONIC. The implementations agree on 39/40 strict decisions in a stratified same-clip panel. Because source corpus, filtering, robot file, and implementation also change, the prevalence difference is not a retargeter comparison. It shows why feasibility should be measured for each corpus-and-pipeline pairing.

This paper contributes:

1) a measured failure diagnosis and policy-independent screen that tests final robot-space trajectories against modeled contacts and actuator limits;
2) an exact-support trial contract with 1,184 units, 368,951 legal starts, explicit terminals, paired conditions, and content hashes; and
3) a manipulation-first comparison of calibrated absolute-learning-progress allocation against deployment-uniform allocation on identical support.

The third item remains a protocol until Phase G completes. A failed manipulation or provenance gate returns `not_tested` without opening a policy endpoint; a valid null would recommend the simpler allocator within the tested task.

II. Related Work

A. Motion filtering and dynamic retargeting

Policy-based curation already removes poorly tracked references. H2O retains AMASS motions that a privileged imitator can follow, and ExBody2 uses an initial policy's sequence errors to select a feasible and diverse subset [3], [9]. These scores combine reference quality with policy capability. KungfuBot instead uses a human-space center-of-mass/center-of-pressure heuristic before robot retargeting [10]. LIMMT combines target-robot motion heuristics, diversity, and complexity, with physical-score weights calibrated through repeated policy training [11]. PHUMA filters source artifacts and retargets with joint-limit, ground-contact, and anti-skating losses [12]. Thus neither filtering nor physics-aware curation is itself new.

Optimization-based retargeters address a complementary problem. Kinodynamic Motion Retargeting uses rigid-body and contact constraints to construct viable locomotion, while Direct Dynamic Retargeting uses simulator-in-the-loop optimization without an infeasible geometric intermediate [13], [14]. AMO and SPIDER likewise generate viable references through trajectory optimization or physics-based sampling with contact guidance [15], [16]. GMR further shows that retargeting artifacts affect downstream tracking robustness [17]. An earlier analytic study estimates velocity, acceleration, and torque requirements for 40 upper-body motions, but does not test whole-body environmental support [18]. Repair can therefore be preferable to exclusion. Our narrower question is diagnostic: after retargeting, can the demanded robot-space wrench be supplied by admissible contacts within modeled actuator limits?

B. Adaptive allocation and evaluation

Prioritized experience and level replay formalize the benefit and bias of nonuniform training distributions [19], [20]. In humanoid tracking, BeyondMimic uses a failure-EMA bin sampler, while GMT and EGM reweight motions or segments using tracking outcomes [5], [7], [8]. Those systems combine allocation with curation, staging, clipping, or architectural changes. Our comparison changes only allocation over a shared set of legal trials.

Robot-learning comparisons are sensitive to seed count, aggregation, and interval construction [21]. Tracking adds a conditioning problem: kinematic error among survivors can improve when a method terminates earlier. HumanTracker similarly finds that kinematic metrics can miss support and contact failures [22]. We therefore use a liveness-weighted primary score and report common-survivor quality only beside survival.

III. Feasibility and Exact-Support Interface

A. Problem and assumptions

Let a retargeted reference provide generalized position and velocity $(q_t, \dot{q}_t)$ for the G1 at frame t. We apply a centered five-frame moving average to velocity, differentiate at the reference frame rate, and smooth acceleration with the same window. The audit is conditioned on a robot model, flat plane, collision geometry, friction, actuator ranges, and reference. It is not a certificate for unmodeled electrical, thermal, compliance, latency, or hardware-safety constraints.

We use the compiled MuJoCo model underlying the tracking environment. Contact-free inverse dynamics returns the generalized force required for the prescribed acceleration. The six free-joint coordinates define the base wrench W_t that the environment must supply; the remaining coordinates define contact-free joint torque τ_t^{free}. A collision geometry becomes a candidate contact when its exact distance to the plane is at most $g = 0.06\,\text{m}$.

The 6 cm band is a declared tolerance. At 3 cm a known feasible control is flagged on 42% of frames because the bank carries an approximately 3 cm ground-alignment offset. At 10 cm the attractor's 7–10 cm floating descent is incorrectly granted a contact candidate. The chosen setting lies between these observed failure modes; it is not a universal constant.

B. Contact-capacity screen

For each candidate point we construct a four-edge pyramidal friction cone with coefficient 0.6 in the primary model; a simulator-matched view makes non-foot contacts frictionless. Mapping nonnegative generator magnitudes f_t through the contact Jacobian yields both base-wrench matrix $A_t f_t$ and joint-torque contribution $J_t f_t$. A weighted nonnegative least-squares problem first exposes the unconstrained residual, with angular-wrench rows weighted by two. We then solve a linear program minimizing ℓ_1 wrench slack subject to

$$-\tau_{\max} \le \tau_t^{\text{free}} - J_t f_t \le \tau_{\max}, \qquad f_t \ge 0. \tag{1}$$

The translational component of the resulting weighted residual defines unsupported force. With no candidate contact, the residual is the contact-free demand.

We report the fraction of frames with no contact candidate (`airborne_frac`), the fraction whose torque-limited unsupported force exceeds half the robot weight (`infeasible_frac`), normalized unsupported-force integral, and joint-torque infeasibility. A clip crosses the primary rule only when `infeasible_frac`> 0.10 (strict inequality). Ballistic flight is not flagged merely for being airborne: near-gravity acceleration has little external support demand. Conversely, nonballistic hovering or descent without support is flagged.

The screen takes approximately one CPU-second per clip in the primary implementation. An independent implementation processes the production bank at 0.145 CPU-s per clip (0.84 ms/frame). These costs support corpus-scale release audit, not real-time control.

C. Exact-support trials

Clip classification is too coarse because one reference can contain both supportable and unsupported intervals. We reduce per-frame screens to ordered feasible runs, retain only 50-step windows wholly inside a run, and bind each motion and sidecar by SHA-256. Across 800 training clips, 1,841 source runs yield 1,184 admissible units and 368,951 legal starts after 657 short runs are discarded. A unit is an attribution stratum; the deployment prior is uniform over legal starts, not units or clips.

The runtime samples only legal starts, emits explicit truncation at a segment boundary, never wraps into a rejected frame, and serializes its deterministic generator and sufficient statistics. Per-unit and per-clip caps bound concentration. Missing sidecars, changed motion hashes, invalid starts, invalid frames, or censored resets fail the contract.

The paired evaluator freezes the motion, start, replicate, environment-noise seed, and dynamics-noise seed across policies. It assigns the reference before the first observation, disables auto-reset until terminal channels are captured, rejects invalid offsets, and records checkpoint, task, condition, reference, evaluator, and output hashes. The 100-clip panel is name- and hash-disjoint from training and contains 2,800 paired conditions.

IV. Experimental Design

A. RQ1: motivating failure

We analyze the previously sealed 100-motion campaign: 4,096 environments, 4,000 PPO iterations, three seeds per arm, and 100 disjoint evaluation motions with eight episodes per clip. It includes failure-adaptive, clip-uniform, and normalized grounded samplers. We report exposure, attractor identity, and held-out survival. This campaign motivates the audit; it is not reused as evidence for the exact-support allocation claim.

B. RQ2: bank-scale screen

The primary corpus contains 10,705 AMASS-derived references retargeted through `whole_body_tracking` to G1. We report raw counts under the fixed rule, category/source breakdowns, and contamination in the historical evaluator. A separate experiment covers all 4,950 references in the filtered BONES-SEED production pairing. We never pool their prevalence. A deterministic 40-clip panel, 20 from each bank and enriched for flags, passes through both implementations to measure score and decision agreement; it is not a prevalence sample.

C. RQ3: negative controls

Simple interventions do not establish training benefit. E-HYG prunes 99 flagged motions from an 800-motion bank but yields a held-out feasible-motion effect of -0.0101 under its predeclared test. Soft segment sampling estimates -0.0196 with a 95% hierarchical-bootstrap interval $[-0.0497, +0.0134]$, but fails its rejected-start-mass manipulation gate. An exploratory segment-native pilot reaches only 0.014 total variation (TV) from control. These negative results motivate an exact-support, manipulation-first test.

D. RQ4: identical-support allocation

Phase G compares two 512-environment, 4,000-iteration arms with confirmation seeds 1–3. G1 samples uniformly over legal starts. G2 estimates each unit's Beta-smoothed conditional success rate $s_u(k)$ and ranks absolute learning progress

$$\mathrm{LP}_u(k) = |s_u(k) - s_u(k-10)|. \tag{2}$$

Its focus distribution is proportional to the legal-start prior times $\mathrm{LP}_u + \lambda$; exploration ρ mixes the G1 prior back in before fixed unit and clip caps.

Before sealing, a 12-row grid crosses $\rho \in \{0.05, 0.10, 0.20, 0.40\}$ and $\lambda \in \{0.001, 0.01, 0.05\}$. Each row runs 50 iterations on seed 20260903. Only sampler ledgers at iterations 30, 40, and 49 enter selection. A row passes when mean TV is in $[0.05, 0.15]$, each checkpoint lies in $[0.025, 0.20]$, entropy-effective units are at least 12, top-1 probability is at most 0.05, invalid/censored counts are zero, and final rank saturation is below 0.90. The passing row nearest TV 0.10 is repeated on seed 20260904. Failure returns the study to design without endpoint access.

After a 400-iteration warm-up, the confirmation gate applies the same separation, concentration, validity, seed, and provenance checks. Only a passed gate unlocks the primary endpoint: liveness-weighted TrackingScore on 25 reference-defined feasible-hard clips,

$$h \exp\left(-\frac{\mathrm{MPKPE}}{0.30\,\mathrm{m}} - \frac{\text{anchor error}}{0.40\,\mathrm{rad}}\right), \tag{3}$$

where h is survived-horizon fraction. The independent unit is clip within training seed, with a seed-then-clip hierarchical bootstrap of 10,000 draws. Survival, all-panel TrackingScore, AULC, and common-survivor non-harm are secondaries. Exactly one status is printed: positive, null, inconclusive, or `not_tested`.

V. Results

A. Failure-adaptive exposure can lock onto a defect

Adaptive peak top-1 mass is 0.884/0.870/0.893 across campaign seeds, and the same BMLmovi kneel-and-crawl clip repeatedly becomes dominant in all three. The maximum belongs to another clip in two seeds. Uniform sampling achieves higher held-out endpoint survival in every paired seed: differences (uniform minus adaptive) are +0.0300, +0.0275, and +0.0300. With only three seeds, the sign pattern is the useful description; the minimum attainable paired permutation p-value is 0.125.

The reference has no joint-limit violation and peak joint speed is at most 5.6 rad/s. The dynamic audit instead localizes the failure (Fig. 1). From 0.75–1.75 s, 86% of frames have no candidate contact and median unsupported force is 329 N; the model weighs 327 N. The 1.75–7.25 s kneeling interval is supportable through non-foot contacts, and the 8.0–8.5 s rise again becomes unsupported. Every trained policy has zero survival from frame zero, while the tested late 8 s offset survives.

B. Prevalence is pipeline-dependent

Under the strict rule, 2,442 of 10,705 primary-pipeline clips are flagged (22.8%). Category rates range from 12.9% for quiet motion to 58.6% for dynamic motion, and source

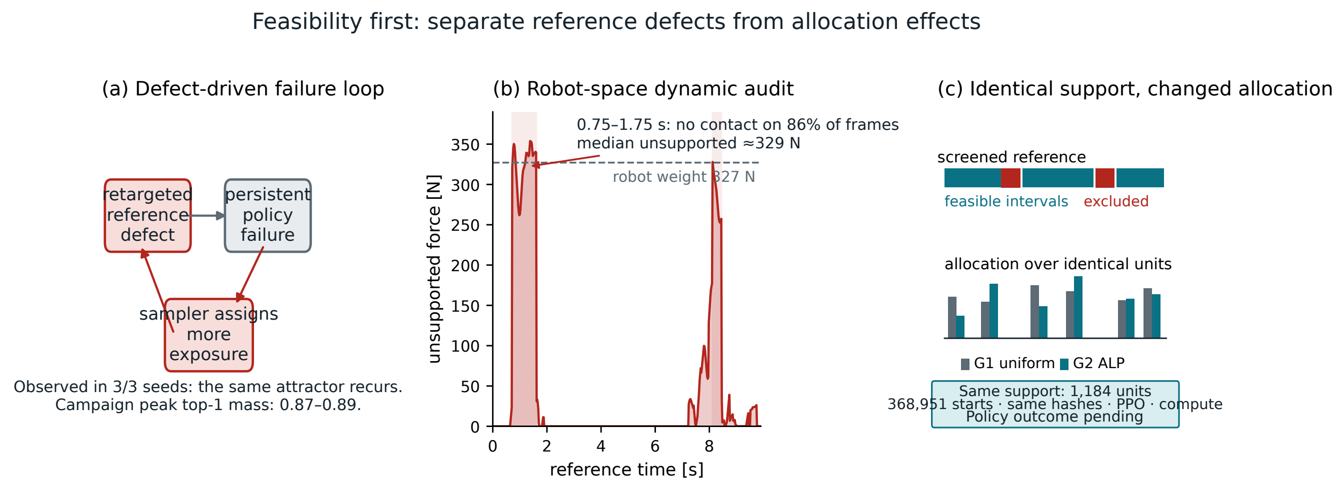

Fig. 1. **The feasibility-first trial interface.** (a) In the motivating simulation campaign, the same attractor recurs in 3/3 adaptive seeds; 0.87–0.89 is the campaign maximum top-1 mass and does not belong to that clip in every seed. (b) A reference-only audit localizes the unsupported interval. This pipeline-specific temporal and modeled-wrench diagnosis does not establish that all adaptive collapse has this cause. (c) The preregistered G1/G2 comparison keeps 1,184 units, 368,951 legal starts, hashes, PPO, and compute fixed while changing allocation. No Phase-G policy outcome is shown.

rates from 0.1% for GRAB to 100% for CNRS. Severity-stratified inspection shows why labels are insufficient: five inspected CNRS cases are ordinary fast walks with extended floating intervals, while the Transitions sample mixes ingest defects with acrobatic content.

Only 7 of 4,950 production clips (0.14%) cross the same nominal rule, while 111 (2.24%) are airborne on more than 10% of frames. Five of seven flags are jumps, four naming a 50 cm box absent from the flat audit scene. Their verdict is *unsupported on this plane*, routing them toward missing terrain or exclusion rather than geometric root projection.

Across the stratified 40-clip panel, `infeasible_frac` rank correlation is 0.9836, `airborne_frac` correlation is 0.9974, and strict decisions agree for 39/40 clips (Cohen's $\kappa = 0.9485$). The disagreement is `burpee_002__A362_M` (0.019 versus 0.136). Figure 2 keeps the full-bank denominators separate and uses only this same-input panel to check implementation agreement; it does not establish corpus or retargeter equivalence.

C. Obvious interventions do not establish allocation benefit

The historical evaluator contains 29 flagged clips among 100. Across three policies, flagged clips score 6.0–8.4 survival points below the all-clip aggregate and 8.4–11.8 below the feasible stratum. The grounded-minus-uniform endpoint contrast is +0.025 on feasible clips and −0.009 on flagged clips. These data justify stratified evaluation, but changing the evaluation subset cannot establish training benefit.

Training-side shortcuts likewise fail to answer RQ4. E-HYG is a sealed null. Soft segment sampling fails its allocation gate. The prior segment-native treatment is only 0.014 TV from control. We retain these failures because they separate three statements often collapsed in discussion: a defect can be measured; screening changes admissible support; and an allocator can improve policy performance on that support. Only Phase G can address the third.

D. Controlled Phase-G allocation result

[Phase G pending: endpoint access remains locked.] The exact 900-motion payload passes all committed identities. The 12-candidate, endpoint-blind calibration is queued behind a 14,000 MiB shared-GPU availability gate. This subsection will be populated only from the frozen result table. If manipulation or provenance fails, the paper will report `not_tested` and no endpoint estimate. If all gates pass, it will report the feasible-hard TrackingScore contrast with seed-then-clip interval, survival, AULC, and common-survivor non-harm.

VI. Limitations and Conclusion

The audit is conditioned on one robot and simulator model and has no hardware closed-loop validation. It models collision contacts, friction, and actuator force limits, but not electrical, thermal, compliance, latency, or structural constraints; *screened feasible* is not a safety certificate. The primary prevalence estimate belongs to one pipeline. The second bank bounds generality but is not a causal retargeter comparison. Both screens assume a flat plane, so terrain-bearing skills require their terrain in the audit.

Finite differences and the contact-distance band introduce modeling error. The 6 cm and half-weight thresholds are design choices supported by local sensitivity checks, not universal constants. Analytic routing also does not establish that exclusion is better than repair. That requires a same-support comparison of certified repaired and excluded intervals.

Phase G has three training seeds, or two under the predeclared budget fallback, so uncertainty is seed- and clip-structured. A positive result would apply only to this ALP rule, support, task, and budget; a null would support G1 only within the same boundary.

Within those limits, the measured conclusion is useful. A failure-adaptive curriculum can repeatedly make a reference whose demanded dynamics have no modeled support source

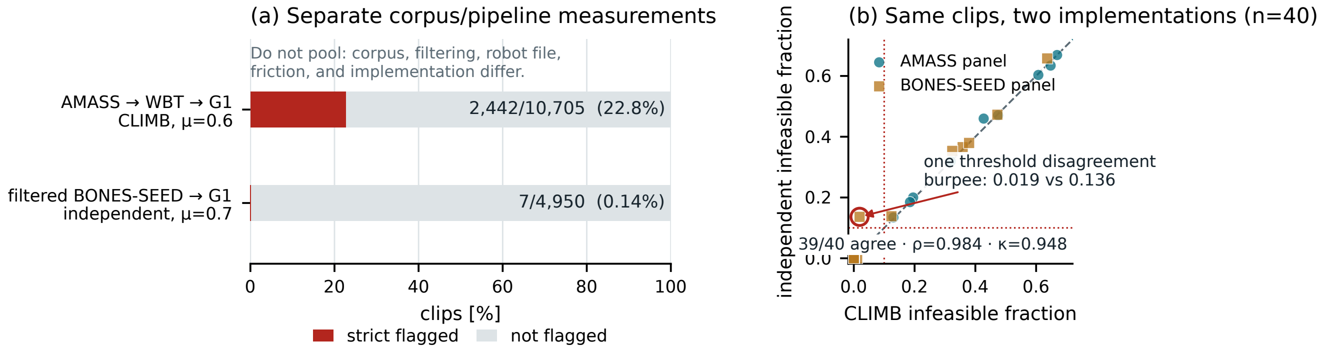

Fig. 2. Bank-scale audit and implementation check. (a) Bars are normalized within two separate corpus/pipeline denominators and apply the strict `infeasible_frac`> 0.10 rule: 2,442/10,705 for AMASS→WBT→G1 and 7/4,950 for filtered BONES-SEED→G1. Corpus, filtering, robot file, friction, and implementation differ, so this is not a causal retargeter comparison. (b) On a flag-enriched, deterministic 20+20 same-clip panel, strict decisions agree for 39/40 clips (Spearman $\rho = 0.9836$, Cohen's $\kappa = 0.9485$). That panel checks implementation agreement; its stratified selection is not a prevalence sample.

a dominant attractor. A policy-independent audit finds such intervals before training, and an exact-support interface prevents feasibility, duration exposure, and terminal semantics from leaking into the allocation comparison. Whether Phase G is positive or null, feasibility must be measured before policy failure is interpreted as learning value.

Reproducibility and AI disclosure

The audit and training stack uses mjlab v1.6.0 at commit `0fb8a681136b`, MuJoCo 3.11.0, NumPy 2.5.1, and SciPy 1.16.2. Motion, sidecar, unit-table, task, checkpoint, evaluator, condition, and output identities are SHA-256 bound. Released code and aggregate artifacts will be supplied through an anonymous review repository; licensed AMASS-derived motion files cannot be redistributed. OpenAI Codex assisted with code, figure composition, and language editing. The authors verified scientific claims and artifact provenance.

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